
Final Report

Introduction to Machine Learning Safety CARLA Autonomous Driving Perception System

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Code repository: [gopesh-007/carla-ml-safety-baseline](https://github.com/gopesh-007/carla-ml-safety-baseline)

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1. System Description

Provided system description

An autonomous vehicle (AV) prototype operates in an urban environment. The system is composed of the following components:

1. **Camera:** A single forward-facing RGB camera, rigidly mounted behind the windshield, provides images at 10 Hz. There are *no other perception sensors* (no radar, no LiDAR, no ultrasonic).
2. **Perception models:** Three separate binary classifiers each receive the camera frame and output one binary prediction: *pedestrian present*, *vehicle present*, or *traffic light present*. Each model was trained exclusively on sunny daytime data collected in the CARLA simulator.
3. **Distance oracle:** A separate model estimates the distance to detected pedestrians and vehicles. You may assume this model is perfect for the sake of this course.
4. **Planning module:** The rule-based planner consumes the three perception outputs together with the current vehicle speed and steering angle. Its primary safety-relevant decisions are whether to continue driving, request deceleration, or issue an emergency brake command. Emergency braking is triggered when the pedestrian detector reports a pedestrian within a critical distance, when the vehicle detector indicates a road user ahead within a speed-dependent threshold, or when the traffic-light detector indicates that stopping is required at an approaching intersection. The planner does not have access to raw camera images and relies entirely on the perception models' outputs.
5. **Vehicle actuators:** Throttle, brake, and steering commands are executed by drive-by-wire actuators. Emergency braking applies maximum deceleration within the physical limits of the vehicle.
6. **Human safety operator:** A trained operator sits in the driver seat and monitors the road. The operator can override the autopilot at any time by pressing the brake pedal or turning the steering wheel. The operator receives a visual dashboard showing the current perception output and system status, but no auditory alerts are provided. During testing, operators work 4-hour shifts.
7. **Operating environment:** The vehicle is tested on public urban roads at speeds up to 50 km/h. The intended conditions are daytime, dry weather, and mapped intersections. However, weather and lighting conditions can change during a test drive (e.g. sudden cloud cover, low sun angle, rain onset).

Key design limitations (by construction)

- No sensor redundancy — a single camera is the only perception input.
- No other safety mechanisms are deployed yet.
- The human operator is the only fallback if the automation fails.
- The operator may exhibit a non-zero probability of delayed or missed intervention, especially under prolonged monitoring.

The traffic-light detector reports only the presence of a traffic light, not its state (red / green). The planner therefore cannot, on its own, decide whether to stop at a signalized intersection. This is a limitation of the system as specified that no model metric can close, so it is recorded as a known design limitation in Section 5.

Architecture & training

All three classifiers use the same ResNet-18 architecture, initialized from scratch without pretrained weights, with the final layer replaced by one logit. Images are resized to 224×224 , converted to tensors, and no data augmentation or pretrained normalization is applied. A sigmoid and threshold of 0.5 are used for evaluation.

For each classification task, the dataset contains 7,200 training, 3,600 validation and 3,600 test frames. Training uses Adam (10^{-3}), batch size 32, weighted binary cross-entropy, and five epochs. The positive-class weights were pedestrian 5482/1718 and traffic light 1924/5276. The evaluated vehicle checkpoint accidentally reused the pedestrian weight 5482/1718, although its class counts are 5,458 positive and 1,742 negative. I retained this historical configuration to keep the published checkpoint reproducible and treat it as a training defect rather than silently changing the experiment. Training was performed on CPU. Code, result files and exact checkpoint hashes are listed in [Appendix A.1](#).

2. Operational Design Domain

The intended ODD is a narrow subset of the provided system’s stated environment. “Detection” below distinguishes a required runtime check from what is currently implemented.

Dimension	Operating conditions	Non-operating conditions	Detection / status
Weather	Dry; clear or partly cloudy; no precipitation	Fog, rain, snow, spray or severe glare	Image/OOD monitor required; k-NN evaluated offline only
Lighting	Daylight with usable contrast	Night, dusk, tunnel or direct low-sun glare	Image-quality/OOD check required; not integrated
Camera	Fixed forward RGB; clean, unblocked, 10 Hz	Blocked, damaged, blurred or missing/late frames	Heartbeat and quality monitor required; absent
Scene type	Mapped urban roads and intersections similar to source town	Unmapped town, highway, unusual road geometry	Map/geofence plus OOD monitor required; absent
Speed	0–50 km/h, forward travel	Above 50 km/h or reverse automation	Vehicle-state check available to planner

ODD gap analysis

Training covers only sunny daytime images from one source domain. It does not cover fog, rain, night, alternative towns, camera faults, glare, or combinations of these factors. The supplied test variants add fog/day/source-town, rain/night/source-town and dry/day/Town-01. These are useful stress tests, but they do not make those conditions part of the validated ODD. In particular, there is no evidence for night in Town-01, fog in Town-01, rainy daylight, or physical camera faults. Performance claims are therefore limited to the standard dry/day/source-town test split.

ODD coverage

The reproducible k -projection calculation uses three recorded context factors: weather, lighting and town. All individual values occur, but their combinations are sparse.

Evidence (k -projection coverage):

k	Observed	Possible	Coverage
1	7	7	100.0%
2	10	16	62.5%
3	4	12	33.3%

Coverage falls sharply as interactions are considered. Consequently, passing a one-factor test cannot justify performance on combined shifts. Uncovered combinations remain unverified residual risk rather than being interpolated from the four observed scenarios.

ODD violation response

The required response is: flag the violation, suppress normal “continue” control, begin a controlled deceleration, issue an unmistakable operator alert, and reach a minimal-risk stop if the operator does not take over within a verified bound. The submitted prototype only evaluates OOD detection offline. It has no integrated OOD signal, automatic minimal-risk manoeuvre, auditory alert, or measured takeover bound. This gap is captured by SC-5, SC-7 and SC-9, evaluated through V-4 and V-5.

3. Safety Analysis (STPA)

Control structure

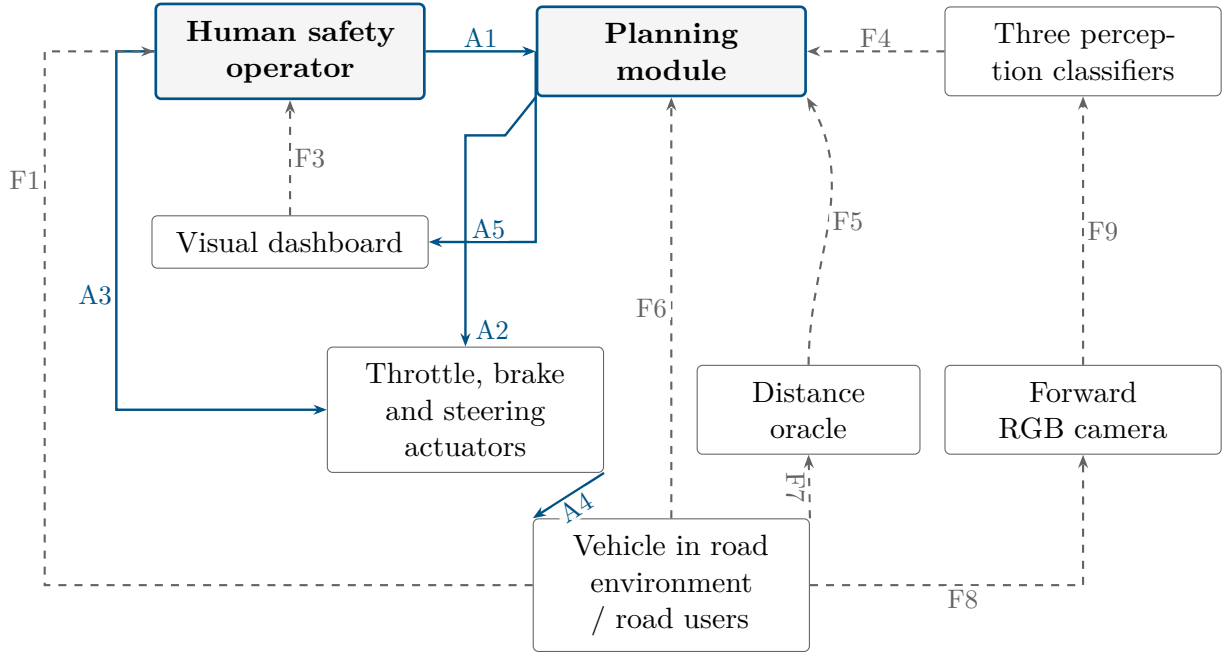


Figure 1: Safety control structure. Solid blue arrows are control actions; dashed grey arrows are feedback.

A1 engage/disengage; A2 continue/decelerate/emergency brake; A3 operator brake/steer override; A4 actuator commands; A5 dashboard status/alert command. F1 direct road view; F3 dashboard display; F4 binary labels; F5 oracle distance; F6 speed/steering; F7 true distance; F8 road scene; F9 camera frames.

The solid arrows distinguish the two control paths. The planning module is the automated controller and issues driving commands through A2, while the human operator controls engagement through A1 and can bypass the planner through the direct override path A3. The actuators translate these commands into physical vehicle motion through A4. The planner also issues dashboard status and takeover alerts through A5.

The dashed arrows close the feedback loops. The planner receives binary perception outputs, oracle distance, speed and steering information through F4–F6. Camera images originate from the road environment through F8 and F9, while the operator receives both a direct view of the road through F1 and the dashboard display through F3.

In STPA, a control action is not inherently unsafe; it becomes unsafe in a particular context. A continue command is normally valid, but becomes unsafe when a pedestrian is within stopping distance, the traffic-light state is unknown, or the vehicle has left its ODD. The planner cannot inspect the image that produced a label, and the operator can override the actuators but receives no auditory alert. These limitations motivate the unsafe control actions below.

Losses and system hazards

ID	Loss	Why unacceptable
L-1	Death or serious injury to occupants or other road users	Irreversible harm to people
L-2	Damage to vehicles, property or infrastructure	Physical and economic harm
L-3	Traffic-law violation	Endangers road users and invalidates road approval
L-4	Loss of availability or traffic continuity	Unnecessary stops can obstruct traffic or provoke rear-end conflict

ID	Hazardous system state	Loss(es)
H-1	Vehicle moves without adequate braking while a pedestrian or vehicle is within stopping distance	L-1, L-2
H-2	Vehicle continues into a signalised intersection without reliable stop/state information	L-1, L-2, L-3
H-3	Automation remains active outside its ODD or with degraded camera input	L-1–L-4
H-4	Planner acts on an incorrect, overconfident perception output without timely override	L-1, L-2
H-5	Vehicle brakes unnecessarily or for too long when no road user requires it	L-1, L-2, L-4

Unsafe control actions

ID	Controller	Control action	UCA type	Hazard	Context in which unsafe
UCA-1	Planner	Emergency brake	Not provided	H-1,H-4	An actual road user is within stopping distance and emergency braking is required
UCA-2	Planner	Brake/decelerate	Too late	H-1,H-2	Action begins after remaining distance is insufficient
UCA-3	Planner	Release brake	Too early	H-1	Road user remains present but a temporary miss/occlusion occurs
UCA-4	Planner	Continue	Provided	H-2	Signal state is unknown at an approaching intersection
UCA-5	Planner	Continue	Provided	H-3	ODD or camera-validity condition has been violated
UCA-6	Planner	Alert/takeover request	Not provided	H-3,H-4	Perception is invalid, OOD or low-confidence
UCA-7	Operator	Engage/keep automation	Provided	H-3	Preconditions are outside the declared ODD
UCA-8	Operator	Override	Too late/not provided	H-1–H-4	Automation fails during prolonged passive monitoring
UCA-9	Planner	Emergency brake	Provided too long	H-5	False positive occurs while no relevant road user is present

Causal loss scenarios

ID	For UCA	Causal scenario
LS-1	1–3	In-distribution false negative or unstable frame-wise output suppresses or prematurely releases braking
LS-2	1–6	Small adversarial perturbation changes a perception output
LS-3	1,4–6	Misclassification is overconfident, so confidence appears safe
LS-4	5–7	Fog, night or new-town input is not detected as outside the validated domain
LS-5	1–3	Single camera is affected by occlusion, small objects, blur or shortcut features
LS-6	4	Traffic-light classifier reports presence but not red/green state
LS-7	6,8	No automatic fallback/auditory alert; four-hour monitoring contributes to delayed override
LS-8	1–3,8	10 Hz acquisition, inference, planning or human response introduces unsafe delay
LS-9	9	Vehicle false positive under fog repeatedly requests unnecessary braking
LS-10	5,7	ODD boundaries are not enforced before or during operation

Safety constraints

ID	Controls	Safety constraint	Level	Verified
SC-1	LS-1,5	ID recall shall be ≥ 0.90 pedestrian and ≥ 0.85 traffic light/vehicle	Model	V-1
SC-2	LS-2	Recall drop at FGSM $\varepsilon = .05$ shall be $< .10$	Model	V-2
SC-3	LS-3	ECE after validation-only calibration shall be $< .05$	Model	V-3
SC-4	LS-4	OOD monitor shall achieve AUROC $\geq .90$ for each tested model/shift	Model	V-4
SC-5	LS-3,4,10	Invalid, OOD or low-confidence input shall suppress “continue” and initiate a minimal-risk response	System	V-5
SC-6	LS-6	Unknown traffic-light state shall never authorize continue; provide state sensing or default stop	System	V-5
SC-7	LS-7,8	Failure shall trigger visual and auditory takeover alert; if no response within a verified bound, stop automatically	System	V-5
SC-8	LS-1,5,9	Brake/release shall use temporal confirmation and hysteresis, with false-positive limits	System	V-5
SC-9	LS-10	ODD preconditions shall be checked before engagement and continuously enforced	System	V-4,V-5
SC-10	LS-8	End-to-end sensing/control latency shall be bounded and braking shall not depend solely on operator reaction	System	V-5

4. Safety Case

Top claim: *The CARLA autonomous driving system operates safely within its Operational Design Domain as specified.*

The claim would require every safety-critical constraint to be supported. The argument below follows $\text{Evidence} \Rightarrow V \Rightarrow \text{SC} \Rightarrow \text{LS} \Rightarrow \text{UCA} \Rightarrow H \Rightarrow L$. A failed verification is retained and propagated as residual risk; it is not reclassified as acceptable.

Verification	SC(s)	Loss scenarios	Hazards
V-1 ID detection performance	SC-1	LS-1, LS-5	H-1,H-2,H-4
V-2 Perturbation robustness	SC-2	LS-2	H-1,H-2,H-4
V-3 Calibrated uncertainty	SC-3	LS-3	H-1–H-4
V-4 OOD detection	SC-4, SC-9	LS-4, LS-10	H-3,H-4
V-5 Safe system fallback	SC-5–SC-10	LS-3,4,6–10	H-1–H-5

V-1: In-distribution detection performance

Constraint: SC-1 requires sufficient ID recall.

Metric: Recall on the 3,600-frame dry/day/source-town test split. Thresholds: pedestrian $\geq .90$, traffic light $\geq .85$, vehicle $\geq .85$.

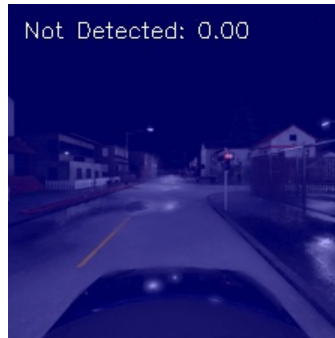
Model	Accuracy	Recall	Threshold	Result
Pedestrian	.7058	.1076	.90	× Not met
Traffic light	.9206	.9358	.85	✓ Met
Vehicle	.7875	.8537	.85	✓ Met

Overall V-1 verdict: × **Not met**

Pedestrian recall is the safety-dominant result: about nine of ten labelled pedestrian-positive frames are missed. Traffic-light and vehicle recall pass individually, although the vehicle result is only 0.0037 above threshold and was produced with the documented class-weight error.



(a) Dry/day, score .08



(b) Night, score .00



(c) Town-01, score .26

Figure 2: Pedestrian Grad-CAM examples across conditions.

The overlays support, but do not prove, the numerical finding. In Fig. 2a the strongest activation lies in distant foliage/sky rather than a pedestrian silhouette. The selected night example has no meaningful activation (Fig. 2b), while Town-01 produces a strong road-edge/structure response (Fig. 2c). This is consistent with shortcut learning and leaves LS-5 open.

V-2: Robustness to input perturbations

Traces to: SC-2 $\rightarrow$ LS-2 $\rightarrow$ UCA-1/4 $\rightarrow$ H-1/H-2/H-4.

Metric: absolute recall drop under untargeted FGSM at $\varepsilon = .05$; required $< .10$.

Model	Clean recall	FGSM recall	Drop	Result
Pedestrian	.1923	.0000	.1923	× Not met
Traffic light	.8987	.0000	.8987	× Not met
Vehicle	.8596	.0351	.8246	× Not met

Overall V-2 verdict: × **Not met**

The deterministic test uses 100 validation images per model (seed 42), so its clean recalls differ from V-1. All drops exceed .10 and pedestrian/traffic-light recall reaches zero. This digital test does not establish a physical attack, but it rejects the stated bound; the non-monotonic vehicle curve at $\varepsilon = .10$ does not repair failure at the specified .05 point.

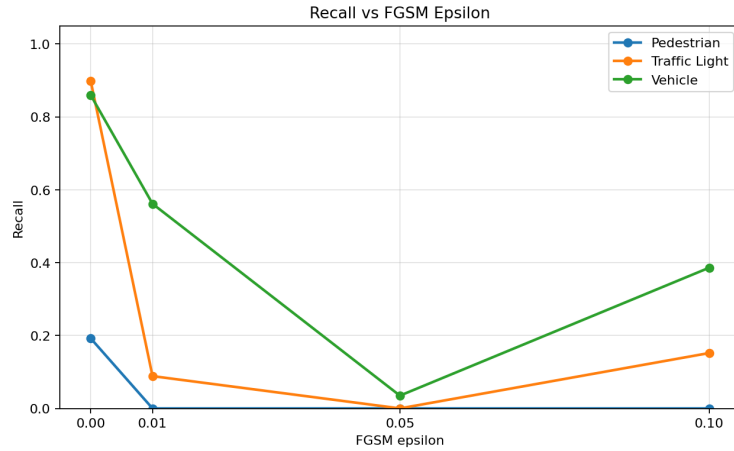


Figure 3: Recall under the saved FGSM evaluation.

V-3: Calibrated uncertainty

Traces to: SC-3 $\rightarrow$ LS-3 $\rightarrow$ UCA-1/4-6 $\rightarrow$ H-1-H-4.

Metric: ten-bin ECE on the test set after temperature selection on validation data; required ECE $< .05$.

Model	Temperature	ECE before	ECE after	Result
Pedestrian	3.0	.1477	.0385	✓ Met
Traffic light	1.2	.0349	.0246	✓ Met
Vehicle	0.9	.0286	.0177	✓ Met

Overall V-3 verdict: ✓ **Met**

Temperature scaling brings all ECE values below .05 without changing class decisions. This supports SC-3 only on ID data: it neither repairs low recall nor implements SC-5's fallback. For the pedestrian model, 3.0 was the upper edge of the search grid; it is therefore the best evaluated value, not evidence that the unconstrained temperature optimum was found.

V-4: Out-of-distribution detection

Traces to: SC-4/SC-9 → LS-4/LS-10 → UCA-5-7 → H-3/H-4.

Metric: feature-space k -NN ($k = 5$) fitted on 1,000 deterministic validation references and scored on an independent 1,000-image dry/day test ID subset against 1,000 OOD images per comparison; required AUROC $\geq .90$ for every tested shift.

Model	Fog	Night	Town-01	Minimum	Result
Pedestrian	.7794	1.0000	.5168	.5168	≈ Partial
Traffic light	.6304	1.0000	.6449	.6304	≈ Partial
Vehicle	.9473	.9875	.4311	.4311	≈ Partial

Overall V-4 verdict: ≈ **Partial**

All three models pass the threshold for night, only the vehicle model passes for fog, and none passes for Town-01. The detector is therefore not reliable enough to justify accepting either fog or a new town. Moreover, the evaluated monitor is an offline experiment and is not connected to the planner. The evidence only partially supports SC-4 and does not satisfy SC-9's continuous enforcement.

V-5: Safe system fallback

Traces to: SC-5-SC-10 → LS-3,4,6-10 → UCA-1-9 → H-1-H-5.

Required design: detected OOD/invalid/low-confidence input causes controlled deceleration, clear takeover request and automatic minimal-risk stop within verified time bounds.

The current design has maximum braking for detected close objects and permits direct human brake or steering override. However, it has no integrated OOD/confidence response, no camera-health monitor, no auditory warning, no verified operator takeover time, and no automatic minimal-risk stop if the operator fails to respond. The presence-only traffic-light output also cannot safely select continue versus stop.

Overall V-5 verdict: × **Not met**

Human supervision reduces some risk but is not a bounded fallback, especially during four-hour monitoring shifts. Since SC-5–SC-10 remain unsatisfied, their associated controls are absent, and V-1/V-2 already expose frequent model failure, the top claim is **not supported**.

5. Residual Risks and Mitigations

From	Open hazard	Residual risk	Concrete next control
V-1	H-1,H-4	Pedestrians are frequently missed in nominal conditions	Retrain on balanced/diverse data; use pretrained detector; validate recall and distance-stratified misses
Stress test	H-5	Vehicle fog false positives can produce unnecessary maximum braking	Correct weighting, include fog negatives, limit false-positive rate, use temporal confirmation/hysteresis
V-2	H-1,H-2,H-4	Small input perturbations can suppress safety detections	Adversarial training, stronger attacks/physical tests, input-integrity monitoring and independent safety channel
V-4	H-3,H-4	Fog and Town-01 shifts can be accepted by the tested detector; all Town-01 comparisons fail	Redesign and tune on held-out ID/OOD data; test more towns and combined conditions; integrate monitor with fail-safe action
V-5	H-1–H-5	Human-only fallback can be delayed or missed	Add audible/haptic alerts, measured takeover bound and automatic minimal-risk stop
ODD cover-age	H-3	Most weather–light–town interactions are untested	Scenario-based sampling for missing 2/3-way combinations before widening the ODD

Known design limitations

- **Traffic-light semantics (H-2):** the classifier I trained answers “is there a traffic light?”, but that turned out to be the wrong question for planning. Presence alone gives no red/green state, so a validated state classifier or infrastructure signal is needed. Until then, an unknown state has to default to stop.
- **Single-camera dependence (H-1,H-3,H-4):** the system has exactly one perception channel. If the camera is occluded or fails, nothing else can compensate. I would add an independent radar/LiDAR channel, together with camera diagnostics that can at least detect a bad input.
- **Classification without localization (H-1,H-5):** a frame-level presence label collapses very different situations—for example, a pedestrian waiting at the curb and one already crossing—into the same signal. Localisation, tracking and path relevance are required before this can serve as a driving perception interface.
- **Offline rather than closed-loop evidence (all hazards):** everything reported here was measured on saved frames. It therefore establishes neither 10 Hz latency nor stopping distance, controller stability, operator response time or collision rate. Closed-loop scenario testing is the necessary next step, not an optional extension of the analysis.
- **Simulator-to-reality gap (H-3,H-4):** I generated no evidence that performance on the supplied simulated images would transfer to real public roads. That gap remains open regardless of how the other model metrics turn out.

The calibrated pedestrian cost experiment (Appendix A.4) shows the trade-off: lowering its threshold eliminates false negatives in this dataset but produces 2,894 false positives. This reduces one hazard by aggravating H-5 and cannot replace a redesigned detector.

Engineering decisions based on the evidence

I used the experiments not only to compare model metrics, but also to decide which claims about the prototype could still be defended. The following decisions were made from the final results.

Preserve the evaluated checkpoints. I did not retrain or replace a checkpoint after seeing the final metrics. In particular, the evaluated vehicle checkpoint reused the pedestrian class weight. A corrected training run might improve the result, but it would no longer represent the experiment completed during the semester. I therefore kept the published checkpoint, documented the defect and treated corrected retraining as future work.

Use safety-relevant metrics rather than the most favourable metric. For detection performance, I gave recall priority because a false negative can suppress a required braking request. The pedestrian model shows why accuracy alone was unsuitable: accuracy was .7058 while recall was only .1076. The opposite problem appeared for the vehicle model in fog. Its F1 score was .8725, but it predicted vehicle-positive in 3,599 of 3,600 frames. I therefore did not use a single favourable aggregate metric to support a safety claim.

Keep the ODD narrow. I did not include fog, night or Town-01 in the validated ODD simply because they had been evaluated. Pedestrian and traffic-light recall reached zero at night, the vehicle model missed 1,748 vehicles, and three-way weather-lighting-town coverage was only 33.3%. These conditions remain stress tests; the supported performance claim is limited to the exact dry/day/source-town test condition.

Use feature k -NN as evidence, not as a deployed safety mechanism. Feature k -NN improved the pedestrian night comparison from .0000 AUROC with MSP to 1.0000, but that benefit did not generalise across conditions. Pedestrian fog AUROC was .7794, its Town-01 result was .5168, and no model passed the Town-01 threshold; only the vehicle model passed fog. I therefore retained k -NN as an experimental monitoring candidate, not as a reliable deployed mechanism or permission for the vehicle to continue driving.

Do not present calibration or threshold reduction as a model repair. I accepted V-3 as evidence that the confidence scores became more reliable, not that the underlying detections became correct. Temperature scaling reduced pedestrian ECE from .1477 to .0385 without changing class decisions. The calibrated cost-sensitive threshold removed pedestrian false negatives on this test set but created 2,894 false positives. I therefore rejected threshold reduction as a standalone deployment solution; retraining, localisation, tracking and temporal decision logic are still required.

Restrict the prototype to research use. The failed pedestrian recall, adversarial recall collapse, incomplete Town-01 OOD detection and absent automatic fallback outweighed the successful calibration result. I therefore restricted the acceptable use to offline analysis, simulation and controlled closed-course research with a safety driver. This leads directly to the deployment recommendation below.

Experimental validity and evidence boundaries

The experiments provide enough evidence to decide the stated verification checks, but they do not cover every condition relevant to vehicle safety. I considered the following limitations when forming the final recommendation.

Frames are not the same as independent driving scenarios. It is tempting to read “3,600 test frames” as a large, statistically robust sample, and I want to resist that interpretation. Neighbouring CARLA frames may come from the same driving sequence and share scenery, weather and road users. I therefore report the number as a frame count—not as 3,600 independent safety scenarios. A proper scenario-level evaluation would need separate runs, defined initial conditions and outcomes such as collision or successful stopping, none of which this dataset provides.

The adversarial result has a narrow scope. I want to be precise about what V-2 actually tested: untargeted one-step FGSM on a deterministic sample of 100 validation images per model, with seed 42. The observed drops are enough to reject the stated bound for this protocol. They do not tell me how frequently adversarial inputs occur in the real world, or how the models behave under PGD, adversarial patches, physical attacks, compression artefacts, camera noise or other attacks. Those are separate questions that this experiment was not designed to answer.

Calibration remains conditional on ID data. I selected temperature on the validation split and computed ten-bin ECE on the test split so that I was not fitting the temperature to the result I would later report. That protects the V-3 result, but only within its dry/day/source-town scope. For the pedestrian model, validation NLL was still decreasing at temperature 3.0, the edge of my search grid. The reported .0385 is therefore the best value I tested, not proof that I found the unconstrained optimum. ECE also depends on the chosen bins, and I did not verify calibration for fog, night, Town-01 or later dataset drift.

The OOD and ODD numbers cover selected factors only. The feature detector used $k = 5$, 1,000 validation references, an independent 1,000-image dry/day test subset as ID, and 1,000 images from each OOD condition. AUROC tells me how well the detector ranks the two groups across a complete comparison; it does not select a runtime threshold or define what the planner should do after a flag. The coverage calculation has a similar limitation. I used weather, lighting and town because those were the factors available in the supplied metadata. Object distance, occlusion, object size, traffic density, glare, blur and camera contamination are absent. The 33.3% three-way result therefore describes the three factors for which I had data, not the complete driving environment.

These thresholds are project criteria, not certification limits. The recall, recall-drop, ECE and AUROC limits are checks defined for this safety case; I am not presenting them as legal or industry-wide acceptance criteria. The asymmetry matters in practice: a failure is enough to leave the linked safety constraint open, whereas a pass supports only the particular metric, dataset and protocol that I evaluated. It does not generalise beyond that evidence.

These limitations do not remove the negative findings. They restrict the scope of the positive evidence and prevent a model-level result from being interpreted as proof of closed-loop system safety.

6. Deployment Recommendation

Recommendation: ☐ Deploy ☐ Deploy with restrictions ☒ **Do not deploy on public roads**

Permitted use: offline analysis and closed-course/simulation research with a safety driver, without presenting the prototype as an autonomous safety function.

Justification: V-1 fails because pedestrian recall is .1076; V-2 fails for all three models; V-4 is only partial; and V-5's fallback controls are absent. V-3 shows that confidence can be calibrated, but calibration does not correct decisions or cause a safe response. The traffic-light state gap and single-camera architecture remain open regardless of model metrics. Therefore the top safety claim is not supported within even the narrow stated ODD. Deployment should be reconsidered only after SC-1–SC-10 are satisfied, the associated controls are implemented, and the complete set is re-verified in closed-loop tests over substantially improved ODD coverage.

The negative recommendation is not a conclusion that the exercises failed. The submitted system is a baseline, and the evidence successfully identifies where its safety argument breaks and which engineering controls are needed before a deployment claim could be made.

Project reflection and next priorities

Looking across the semester, the most useful lesson for me was that a favourable model metric is not automatically a favourable safety result. The pedestrian model reached .7058 accuracy while recall was only .1076, and the vehicle model reached .8725 F1 in fog while predicting positive in 3,599 of 3,600 frames. Calibration passed its own verification without changing a classification, and feature k -NN separated night well but failed several fog and Town-01 comparisons without providing a safe planner response. These results only became meaningful when they were connected to unsafe control actions and possible losses.

The same distinction applies to the safety mechanisms. Calibration, OOD detection and Grad-CAM were useful diagnostics, but none of them can reduce risk unless the system reacts appropriately. An uncertainty or OOD flag that still allows “continue” does not close the hazard. The required chain is detection, a bounded planner response, a clear operator warning and an automatic minimal-risk stop if takeover does not occur.

If I continued the project, I would work in the following order:

1. **Rebuild the perception baseline:** improve pedestrian detection on nominal data, correct the vehicle class weight, and provide traffic-light state rather than presence alone.
2. **Implement the missing system response:** add camera-health checks, connect calibrated uncertainty and OOD monitoring to the planner, and implement the verified fallback and alert path.
3. **Generate closed-loop evidence:** test stopping distance, end-to-end latency, operator takeover and the missing weather–lighting–town combinations with predefined acceptance criteria.

Adversarial training, stronger attacks and further explainability work would still be useful, but I would not prioritise them above failed nominal pedestrian detection and the absent fallback. The main outcome of this project is therefore not a claim that the prototype is safe, but a clearer account of the evidence and engineering controls that would be required before that claim could be responsibly reconsidered.

Statement of Authorship

AI-use disclosure. I used OpenAI Codex as a supporting tool for code review and debugging, organising and cross-checking experimental evidence, and assisting with drafting, revising and formatting parts of the report in LaTeX. I ran the experiments, verified the reported evidence, and made the final analytical and safety decisions. All AI-assisted suggestions were critically reviewed and revised before inclusion.

I hereby declare that I have independently written the present work and have not used any sources or aids other than those stated.

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Excluded from the requirement of identification are orthographic or grammatical corrections, translations, and improvements to formulations that do not alter meaning.

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I confirm that I have fully understood the content of my work and can represent it independently.

Magdeburg,
19th August 2026
Place, Date


Signature

A. Additional Material

Reproducibility record

The public repository is <https://github.com/gopesh-007/carla-ml-safety-baseline>. Exact evaluated checkpoints are published in release `v1.0-model-checkpoints`. SHA-256 values are:

File	SHA-256
<code>pedestrian_model.pth</code>	033e991fed2412c7675aa25aada36194d2505c722636fe2b44d5fe67d8e01e0a
<code>traffic_light_model.pth</code>	7a2444f42eb8d20c0d515396945eda35348952ab1d7d06329503562547929650
<code>vehicle_model.pth</code>	f6d98f8c1bb08fd7201d0a435c032655932d01b0174a75440827d99992f04d2a
<code>pedestrian_model_backdoor.pth</code>	cd13f2d3d266e485b8434a000fea883ded5ce68f6b4fd28f518920b3f126b4bb

The repository file `EVIDENCE.md` maps each final verification to its script, checkpoint, protocol and saved output. Principal evidence files are:

- `outputs/adversarial/adversarial_results.csv`
- `outputs/uncertainty/calibration_results.csv`
- `outputs/ood/knn_all_models_results.csv`
- `outputs/ood/odd_k_projection_coverage.csv`

The dataset was supplied for the course and is not redistributed.

Evidence provenance

The table below is the compact provenance record for the five final verifications. It distinguishes saved experimental evidence from V-5, which is a negative design inspection rather than a missing experiment. Exact commands, checkpoint hashes and supporting artifacts are recorded in `EVIDENCE.md`.

ID	Implementation	Evidence record	Protocol	Verdict
V-1	<code>evaluate_pedestrian.py</code> ; <code>evaluate_traffic_light.py</code> ; <code>evaluate_vehicle.py</code>	Evaluator console metrics recorded in the README and Section 4	Complete dry/day/source-town test split; 3,600 frames per model; threshold .5	× Not met
V-2	<code>evaluate_adversarial.py</code>	<code>outputs/adversarial/adversarial_results.csv</code>	Random validation subset; 100 frames per model; seed 42; FGSM $\epsilon = .05$	× Not met
V-3	<code>evaluate_uncertainty.py</code>	<code>outputs/uncertainty/calibration_results.csv</code>	Temperature chosen on validation; ten-bin ECE on all 3,600 test frames	✓ Met
V-4	<code>evaluate_ood_knn_all_models.py</code>	<code>outputs/ood/knn_all_models_results.csv</code>	$k = 5$; 1,000 validation references; independent 1,000-image test ID and 1,000 OOD images per comparison	≈ Partial
V-5	System-description and STPA inspection; no fallback implementation	Sections 1, 3 and 4; no generated result file	Required fallback functions checked against the described design	× Not met

The V-1 evaluators predate the structured CSV records, so their terminal results are explicitly identified as transcribed evidence rather than being represented as a file that does not exist. The README also retains provisional STPA notation from earlier exercises; the final SC/V identifiers in this report are authoritative for submission.

Adverse-condition performance

Model	Condition	Accuracy	Precision	Recall	F1	Interpretation
Pedestrian	Fog	—	—	.315	.269	partial degradation
Pedestrian	Night	—	—	.000	.000	complete miss failure
Pedestrian	Town-01	—	—	.277	.180	below safe threshold
Traffic light	Fog	—	—	.000	.000	complete miss failure
Traffic light	Night	—	—	.000	.000	complete miss failure
Traffic light	Town-01	—	—	.283	.416	severe degradation
Vehicle	Fog	.7739	.7738	1.0000	.8725	3,599/3,600 frames positive
Vehicle	Night	.4517	.8211	.3724	.5124	1,748 false negatives
Vehicle	Town-01	.6486	.6885	.7516	.7187	domain-shift degradation

For vehicle/fog, the confusion counts are TP 2,785, FP 814, TN 1 and FN 0. The high recall and F1 therefore hide near-constant positive prediction, illustrating why a single favourable metric cannot discharge H-5.

Additional exercise evidence

Cost-sensitive pedestrian decision. With false negatives costed at 100 and false positives at 1, the uncalibrated .5 threshold produced 630 FN, 429 FP and total loss 63,429. After temperature scaling, the Bayes threshold $1/(100 + 1) = .0099$ produced 0 FN and 2,894 FP (loss 2,894). This is safer with respect to missed pedestrians but operationally unusable without localisation and temporal logic.

Backdoor experiment. A fixed trigger was applied to 171 of 1,718 positive pedestrian training frames (10% of that class, 2.4% of the complete training split). The backdoored checkpoint achieved 100% attack success rate on triggered test inputs. This evidence is not part of V-2's FGSM threshold, but it supports the broader need for training-data provenance and integrity checks.

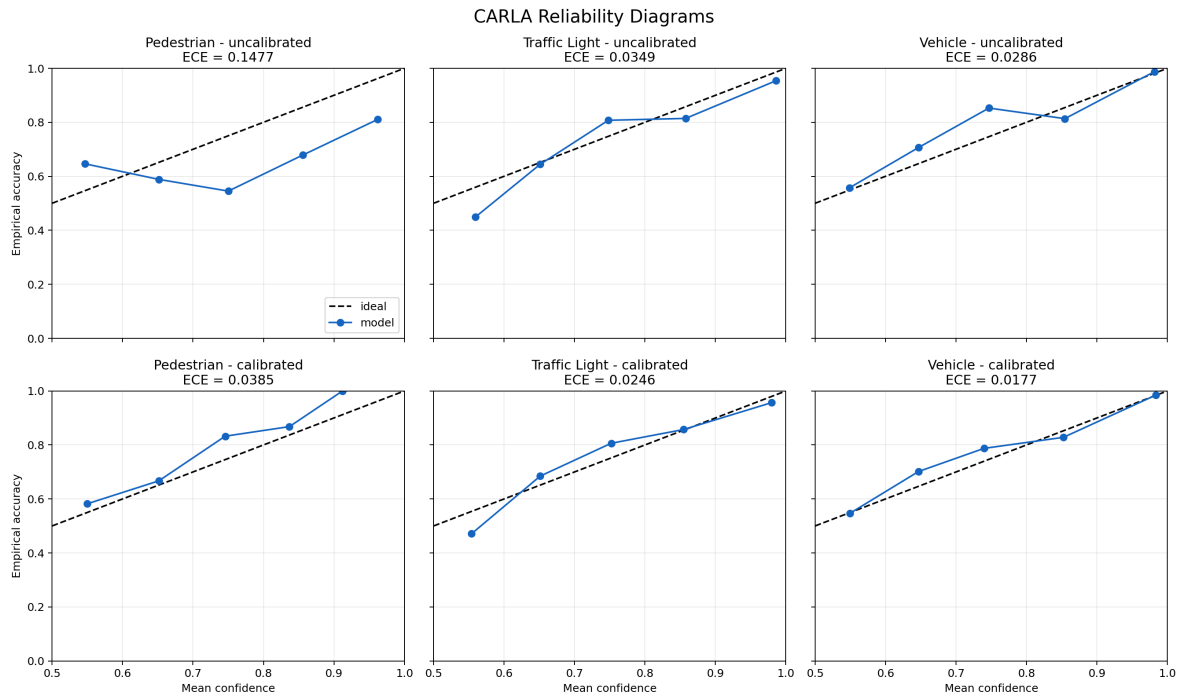


Figure 4: Reliability diagrams before and after temperature scaling.

Evolution of the STPA analysis

The STPA in Section 3 was not written only at the end of the project. I began with the control structure and loss scenarios from Exercise 2, then updated the analysis whenever a later experiment exposed a new failure mechanism. The table below records the changes that are supported by the saved evidence or by assumptions in the final system specification.

Stage	Evidence or system assumption	Change made to the STPA
Initial Exercise 2 analysis	The first control structure identified missed braking, operation outside the ODD, unknown traffic-light state and delayed human intervention as central risks.	Established the original losses, hazards and UCAs. These remained the basis of the final analysis rather than being replaced by a separate assessment. Trace: L-1–L-4; H-1–H-4; UCA-1–UCA-8
Adversarial-ML exercise	At FGSM $\varepsilon = .05$, recall dropped by .1923, .8987 and .8246 for pedestrian, traffic-light and vehicle models respectively.	Added adversarial perturbation as a causal explanation for missing or incorrect perception, together with a measurable robustness constraint. Trace: LS-2; SC-2; V-2; H-1,H-2,H-4
Uncertainty exercise	The pedestrian model was strongly miscalibrated before scaling (ECE .1477); temperature scaling reduced it to .0385, but did not change the underlying classifications.	Added overconfident misclassification as a loss scenario and separated model calibration from the still-missing system response to low confidence. Trace: LS-3; SC-3,SC-5; V-3

STPA evolution (continued)

Stage	Evidence or system assumption	Change made to the STPA
OOD and ODD exercises	Feature k -NN separated night for all models, but only the vehicle model met the fog threshold and no model met it for Town-01. Three-way ODD coverage was only 33.3%.	Added undetected domain shift and unenforced ODD boundaries. Detection and safe response were kept as separate constraints because the evaluated monitor is offline. Trace: LS-4,LS-10; SC-4,SC-9; V-4; H-3,H-4
Vehicle robustness evaluation	In fog, 3,599 of 3,600 frames were classified as vehicle-positive, including 814 false positives.	Added the possibility of an unnecessary or prolonged braking request and a constraint for temporal confirmation, hysteresis and false-positive control. Trace: H-5; UCA-9; LS-9; SC-8
Final system specification	The report specification assumes a perfect distance oracle, while the traffic-light model still reports presence rather than signal state.	Removed distance-underestimation as a model failure scenario from the earlier analysis, but retained the unresolved traffic-light-state scenario and its default-stop constraint. Trace: LS-6; UCA-4; SC-6; H-2

I did not introduce a new hazard for every experiment. Where possible, later findings were linked to an existing unsafe control action and expressed as a more specific causal scenario or safety constraint. This kept the system-level hazard definition stable while allowing the evidence and mitigations to become more precise over the semester.